

The image features the text "DL for ENG" in a bold, 3D, gold-colored font. The letters are floating on a vast, blue ocean with visible ripples and small waves. In the background, there are rolling blue hills or mountains under a sky filled with soft, grey clouds. The overall scene has a serene and expansive feel.

DL for ENG

## LECTURE 10.1

# Ionic Diffusion

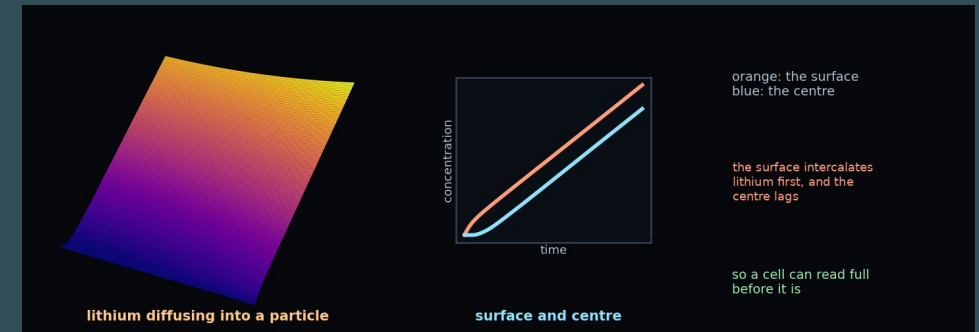
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Deep Learning for Engineering • 2026

### PARTICLE • VOLTAGE • INVERSE



*Lithium entering one particle: the surface intercalates lithium first and the centre lags, because lithium crosses only the surface and diffuses inward slowly. The lecture names the cell model's five fields - lithium in particles and electrolyte, two potentials, temperature - then solves the particle, reads the state of charge, and recovers the diffusivity from the surface.*

# Learning Objectives

## SECTIONS, IN ORDER

- 1 The cell** — five fields, two scales, and heat both ways
- 2 The particle** — diffusion, scaling, state of charge
- 3 PINN and inverse** — the start built in, the diffusivity found

*Incropera carries the transport and the diffusion, Liu the inverse method; Gu & Wang (2000) the coupling, Chen et al. (2020) the LG M50 parameters. The example is Ex\_10.1's graphite particle; L10.2 takes the same physics to a solid oxide cell.*

## READ, AND WHAT EACH ANSWERS

**Incropera ch. 1** which transports share one law?

**Incropera ch. 14** how does lithium reach the surface?

**Liu ch. 4** what can readings identify?

# Recap of L9.2 Gas Flow

## 1 – 2 · GAS AND SPECIES

### 1 Which three facts make the gas in a fuel channel a known, incompressible flow?

Its Mach number is below a thousandth, so the driving pressure drop is far too small to change the density. The electrode takes one  $\text{H}_2$  and returns one  $\text{H}_2\text{O}$ , so the moles per volume and the velocity stay, though the mass density rises. At  $\text{Re } 1.3$  the entrance is about a millimetre, so the parabola holds from the inlet. *White ch. 9; Ex\_09.2 section 3*

### 2 Why can diffusion along the channel be dropped, and what does that make the equation?

The Péclet number along it is 125, so that term is a ten-thousandth of the one across. Without it the equation is parabolic in  $x$ : it marches from the inlet like heat in time and needs no outlet condition. A zero-slope outlet would be wrong, as the mean keeps rising. *Incropera ch. 14; Ex\_09.2 sections 3 – 4*

## 3 – 4 · ONE PINN, EVERY SPEED

### 3 What does the PINN's trial function build in, and what does a loss term have to hold?

The hand calculation  $X/s$ , so the network learns only the profile across, in units of  $G$ . A loss term holds the network's flow-weighted mean at zero, or the balance drifts. The Sherwood number is a tenth of a per cent of the field: read wall and mean from the same column, and both methods find it to 0.3 %. *Incropera ch. 8; Ex\_09.2 section 5*

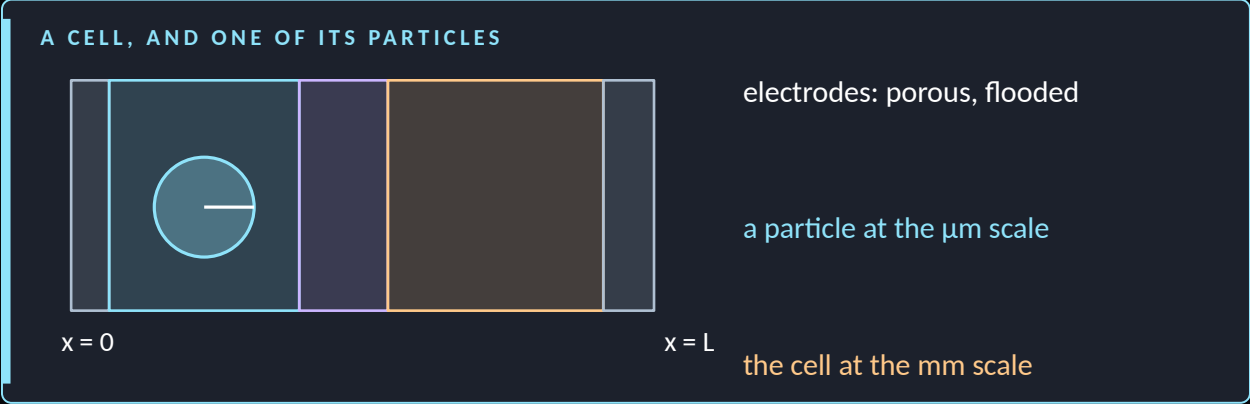
### 4 What can a PINN with the speed as an input do that a march cannot?

Give the field at any speed without solving again, and its derivative with respect to the speed by autograd. Newton's method then finds the speed for 25 % hydrogen at the outlet, 0.3174 m/s, the exact value, with no sweep. A new speed costs one evaluation; a march starts again. *Liu ch. 2.2; Ex\_09.2 section 6*

# Inside the Cell

**Lithium-ion cell** — a sandwich of two porous electrodes around a separator, flooded with electrolyte. Lithium lives in two places at once: inside the solid particles, and dissolved in the liquid between them. A model of it solves five fields on two length scales.

## THE SANDWICH, AND THE SCALE INSIDE IT



The cell is millimetres across; a particle is micrometres. Lithium must diffuse from each particle's interior to its surface before it can leave, so the model has a coordinate inside a coordinate.

## THE FIVE FIELDS OF A CELL MODEL

| field                       | lives    | equation  |
|-----------------------------|----------|-----------|
| $c_s$ in the particles      | 0 to $R$ | parabolic |
| $c_e$ in the electrolyte    | 0 to $L$ | parabolic |
| $\phi_s$ in the solid       | 0 to $L$ | elliptic  |
| $\phi_e$ in the electrolyte | 0 to $L$ | elliptic  |
| $T$ , the temperature       | the cell | parabolic |

Every type in the table is one this course has solved: L8's heat equation and its Poisson problems, wearing electrochemical names. What is new is that the five supply terms to each other. Ex\_10.1 solves the first row; the rest is this lecture's.

# Two Phases Two Potentials

**Charge conservation** — two elliptic equations, one per phase, with no time derivative: at every instant the potentials adjust to carry the current. The reaction current appears in both with opposite signs, because charge that leaves the solid enters the electrolyte.

## THE TWO EQUATIONS, AND THEIR SHARED SOURCE

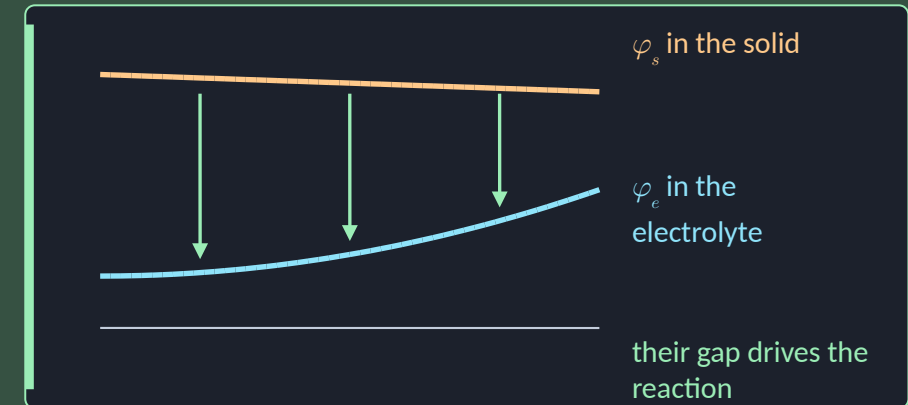
$$\nabla \cdot (\kappa^{eff} \nabla \phi_e) + \nabla \cdot (\kappa_D^{eff} \nabla \ln c_e) + j^{Li} = 0$$

$$\nabla \cdot (\sigma^{eff} \nabla \phi_s) - j^{Li} = 0$$

$\phi_e, \phi_s$  the potentials of electrolyte and solid,  $\kappa^{eff}, \sigma^{eff}$  their effective conductivities  
 $c_e$  the lithium in the electrolyte,  $j^{Li}$  the reaction current per volume, opposite signs

**TALK • 2 MIN** How would you explain, in your own words, what an equation with no time derivative says about its field? Then, in the drawing on the right, what happens to the green arrows where the two curves run parallel?

## ACROSS ONE ELECTRODE



**the extra term** a gradient of  $c_e$  drives  $\phi_e$

**why** the two ions move unequally

*The gap  $\phi_s - \phi_e$  drives the reaction.*

# Three Model Levels

**Model hierarchy** — the three levels of a cell model: the single particle model (SPM) keeps one particle per electrode, the SPMe adds the electrolyte, the DFN keeps a particle at every point of the electrode. The level is chosen by the current, not by ambition.

## PICK THE LEVEL BY THE CURRENT

| model | solves                            | holds to |
|-------|-----------------------------------|----------|
| SPM   | two particles, one current        | about 1C |
| SPMe  | adds $c_e$ and $\phi_e$ along $x$ | about 3C |
| DFN   | a particle at every node of $x$   | any rate |

For a PINN the choice costs more than for a classical solver: each added field is another network, another residual, another loss term to balance. A matrix only grows.

## WHAT THE EXERCISE KEEPS

|                    |                               |
|--------------------|-------------------------------|
| <b>the field</b>   | one particle's $c_s(r, t)$    |
| <b>the current</b> | shared evenly: $j$ known      |
| <b>the voltage</b> | algebra after the solve       |
| <b>left out</b>    | electrolyte, potentials, heat |

*In the single particle model only the diffusion is a differential equation. Charge conservation is trivial - identical particles share the current evenly - and the voltage is one line of algebra once the particle is solved. MP10.1A takes the full DFN on.*

# Heat Runs Both Ways

**Thermal coupling** — the loop between electrochemistry and temperature: the reaction, its entropy and the resistances make heat, and the temperature changes the diffusivities and rates that make it. A model with properties held at one temperature gets the trend backwards.

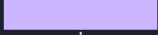
## THE ENERGY EQUATION, AND ITS THREE SOURCES

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + q, \quad q = q_{rxn} + q_{rev} + q_{ohm}$$

$\rho c_p$  the heat capacity per volume,  $k$  the thermal conductivity,  $q_{rxn}$  reaction,  $q_{rev}$  entropic,  $q_{ohm}$  ohmic

### THE THREE SOURCES

reaction  $q_{rxn}$   two terms only heat

entropic  $q_{rev}$  

ohmic  $q_{ohm}$  

cools heats

one changes sign with the current

## WHAT GU & WANG FOUND

**coupled model** warming helps: transport improves

**decoupled model** sees the heating, not the help

**the result** the trend backwards

**in a large cell** tens of degrees across it

**for a PINN** one more residual term

*The entropic coefficient was long unmeasured; O'Regan et al. (2022) supply it for the LG M50.*

# Diffusion Inside a Particle

**Solid diffusion** — Fick's second law in a sphere: L8.2's heat equation in spherical coordinates, with symmetry at the centre and the reaction's flux at the surface. The reaction sees the surface concentration, not the average, so a particle can read as empty before it is.

## THE EQUATION, AND ITS TWO CONDITIONS

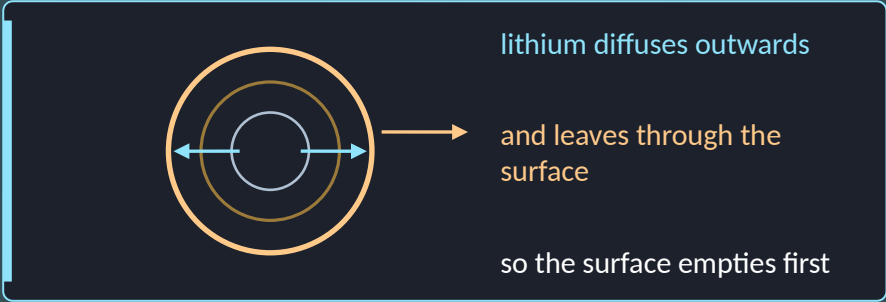
$$\frac{\partial c_s}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 D_s \frac{\partial c_s}{\partial r} \right)$$

$$\left. \frac{\partial c_s}{\partial r} \right|_{r=0} = 0, \qquad -D_s \left. \frac{\partial c_s}{\partial r} \right|_{r=R} = j$$

$c_s$  the lithium in the particle,  $r$  the radius,  $D_s$  the solid diffusivity,  $R$  the particle radius,  $j$  the surface flux

**TALK • 2 MIN** How would you explain, in your own words, why the reaction sees the surface and not the average? Then, in the drawing on the right, what happens to the gap between surface and centre when the current doubles?

## EX\_10.1'S GRAPHITE PARTICLE



|                                                                                                                        | value                                      |
|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| radius $R$                                                                                                             | 5.86 $\mu\text{m}$                         |
| diffusivity $D_s$                                                                                                      | $3.3 \times 10^{-14} \text{ m}^2/\text{s}$ |
| diffusion time $R^2/D_s$                                                                                               | 1041 s                                     |
| discharge                                                                                                              | 1C: 5 A for an hour                        |
| <i>Chen et al. (2020), as PyBaMM distributes them. The hour and the diffusion time are alike: the profile matters.</i> |                                            |

# Scaling the Problem

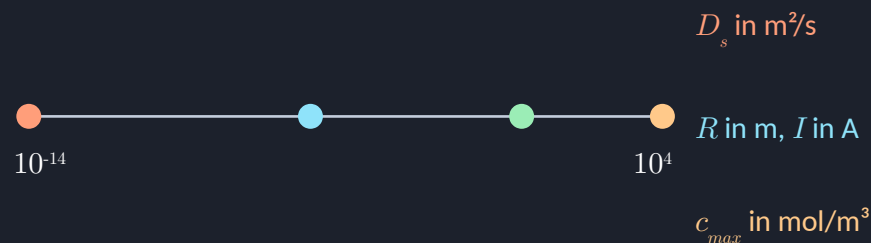
**Scaling** — measuring each quantity in a unit the problem itself supplies, so every number the network meets is of order one. In raw SI units these quantities span about twenty orders of magnitude, one residual term dwarfs the rest, and no loss weight makes up for that.

## THE SCALED PROBLEM, AND ITS ONE CONSTANT

$$u_\tau = C(u_{rr} + \frac{2}{r}u_r), \quad C u_r = 1 \text{ at } r = 1, \quad C = \frac{D_s t_{end}}{R^2} = 3.46$$

$u$  the lithium lost, over  $c_{ref} = j t_{end}/R$ ,  $r$  over  $R$ ,  $\tau$  over the hour  $t_{end}$ ,  $C$  the scaled diffusivity

### THE RAW QUANTITIES, ONE LOG SCALE



## THE SCALES, SUPPLIED BY THE PROBLEM

|               |                         |
|---------------|-------------------------|
| length        | the radius $R$          |
| time          | the hour $t_{end}$      |
| concentration | $c_{ref} = j t_{end}/R$ |
| what is left  | one number, $C = 3.46$  |

One free gift of these units: the mean of  $u$  over the sphere is exactly  $3\tau$  - what has left is the current times the time - whatever the diffusivity. The mass balance, and section 5 puts it in the loss.

# State of Charge and Voltage

**State of charge** — how full the cell is. The bulk value is the mean concentration: a coulomb counter, exactly  $1 - \tau$  here, needing no field. A gauge that has lost count must read the voltage, and the voltage sees only the surface - it is the solved field that converts the one into the other.

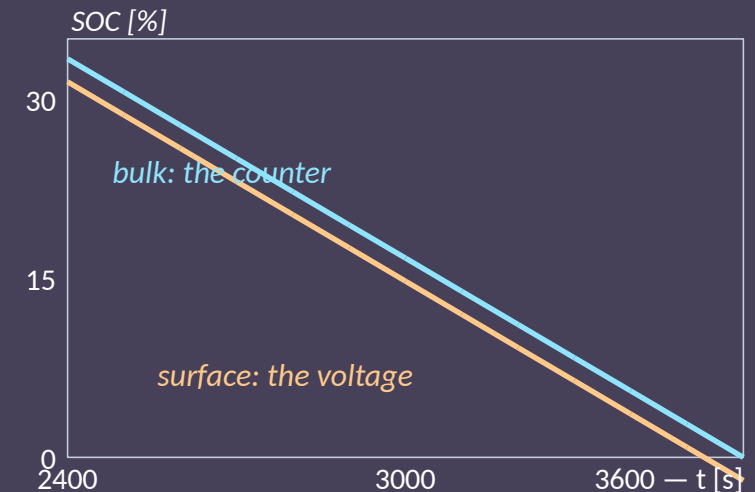
## THE SURFACE SETS THE VOLTAGE

$$\theta = \frac{c_{se}}{c_{s, \max}} \quad V = U_p(\theta_p) - U_n(\theta_n) + \eta_p - \eta_n$$

$\theta$  the surface state of charge,  $c_{se}$  the surface concentration,  $c_{s, \max}$  the most a particle holds  $U_p, U_n$  the measured open-circuit curves,  $\eta_p, \eta_n$  the overpotentials (left out in Ex\_10.1)

The open-circuit curves are measured, not derived: strongly nonlinear, with plateaus where two phases coexist. On LFP the plateau moves some 10 mV over most of the charge, so there the voltage says almost nothing about the surface - Ex\_10.1's film shows it.

## THE TWO GAUGES, OVER THE HOUR



Ex\_10.1's exact field, the last 20 minutes: the surface reads about 2 points low and crosses zero a minute early - the cell cuts off with charge still in the bulk.

# A PINN for the Particle

**Trial function with the start built in** — a network wrapped so the initial condition holds whatever the weights do:  $\tau$  times the network is zero at the start, as L8.2's pulse was. The residual is multiplied by  $r$ , so it has no  $1/r$  in it and at the centre it reduces to the symmetry condition for free.

## THE TRIAL FUNCTION, AND THE BALANCE IN THE LOSS

$$\hat{u}(r, \tau) = S \tau \mathcal{N}(r, \tau), \quad \bar{\hat{u}} = 3\tau$$

$S = 3$  the size of the answer,  $\tau \mathcal{N}$  zero at the start whatever the network does,  $\bar{\hat{u}}$  its mean over the sphere

### THE TRIAL FUNCTION IN PYTORCH

```
net = MLP(n_in=2, n_hidden=32, n_layers=4) # r and tau in
u = 3.0 * tau * net(p) # the trial function: zero at the
start
g = grad(u, p) # first derivatives by autograd
res = r * g[:,1:2] - C * (r * d2(u, p, 0) + 2 * g[:,0:1]) # the residual, times
mass = (u(q_pts) * q_w).sum(1) - 3 * tau_q # the mass balance, held at 0, w
= 1000
```

## THREE METHODS, ONE TRUTH

|                              | FVM                    | PINN                   |
|------------------------------|------------------------|------------------------|
| surface at 3600 s            | 890 mol/m <sup>3</sup> | 899 mol/m <sup>3</sup> |
| worst error                  | 29 mol/m <sup>3</sup>  | 21 mol/m <sup>3</sup>  |
| the hour, online             | a few ms               | under 1 ms             |
| training, offline            | none                   | about 45 s             |
| agreed 30 mol/m <sup>3</sup> | 40 shells              | met                    |

*The exact series is the truth: surface 891, centre 2260 mol/m<sup>3</sup> at the end. The mass balance must be a loss term: a network a little late with the flux at the start loses lithium it never gets back. The surface points exclude  $\tau = 0$ , where the trial function has no gradient.*

# The Diffusivity From the Surface

**Identifiability** — whether the data can tell the parameter at all. The voltage reads the surface through the open-circuit curve, and the surface's lag behind the mean is what carries the diffusivity. On LFP's plateau that reading goes nearly blind.

## ONE MORE TRAINABLE NUMBER

$$C \text{ trainable, } \mathcal{L} = \mathcal{L}_{PDE} + w_S \mathcal{L}_{flux} + w_M \mathcal{L}_{mass} + w_D \mathcal{L}_{data}$$

$\mathcal{L}_{data}$  the misfit to 37 surface readings,  $w_S = 10$ ,  $w_M = 1000$  hold the flux and the mass balance

### THE INVERSE PROBLEM IN PYTORCH

```
logC = torch.nn.Parameter(torch.log(C0))    # the diffusivity, trained with
the weights
C = torch.exp(logC)                        # positive by construction
L = res(C).pow(2).mean() + W_D * (u_surf - readings).pow(2).mean() # physics +
data
optimizer.backward(); opt.step()           # one step moves weights and C
together
```

## WHAT EX\_10.1 RECOVERS

**the data** 37 surface readings

**the noise** 50 mol/m<sup>3</sup> on each

**the start** half the true value

**found**  $D_s$  0.38 % low, in a minute

**not divided by  $C$**  divided,  $C$  runs away

*Unlike Ex\_08.2: divided by  $C$ , a uniform particle fits the readings and  $C$  grows without limit.*

# Advanced Topics

## RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

| topic                          | what it adds                                    | read                  | where it makes a difference       |
|--------------------------------|-------------------------------------------------|-----------------------|-----------------------------------|
| the electrolyte, the SPM       | lithium and its potential between the particles | Incropera ch. 14      | above about 1C                    |
| <u>Butler–Volmer kinetics</u>  | the exponential rate at the interface           | PyBaMM docs           | the overpotentials in the voltage |
| <u>the full model, the DFN</u> | a particle at every point of the electrode      | Marquis et al. (2019) | above about 3C; MP10.1A           |
| heat in two dimensions         | the core runs hotter than the can               | Incropera ch. 5       | large cells and packs             |
| <u>ageing and lost lithium</u> | slower diffusion, capacity gone for good        | O'Regan et al. (2022) | the BMS; MP10.1B                  |

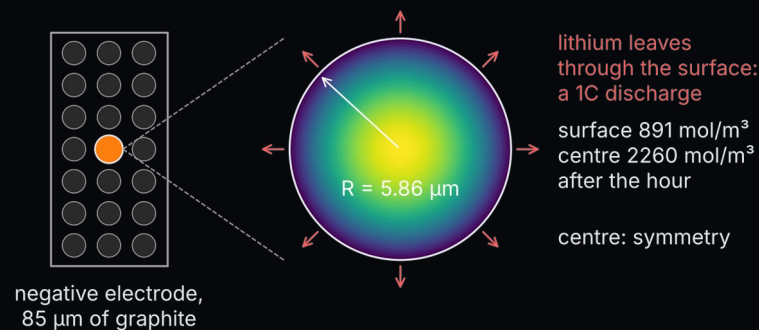
### WHY THEY ARE LEFT OUT

Ex\_10.1 solves one particle with a known flux, scored on an exact series, so none of them is needed to do it. In the single particle model the current is shared evenly and the voltage is algebra; the rows return in the two mini projects.

# Exercise 10.1

## ONE NOTEBOOK, SECTION BY SECTION

- 1 - 3 — the particle, its data, its scales, the mass balance
- 4 — the exact series; finite volumes; the gauges and the film
- 5 — a PINN with the start built in; the three compared
- 6 — the diffusivity from 37 noisy surface readings
- 7 - 9 — the report; mini projects MP10.1A and MP10.1B



## THE QUESTION EACH PART ANSWERS

- 1 - 3 which scales make it order one?
- 4 what does the surface hide?
- 5 can the PINN match the grid?
- 6 what do readings identify?

## REPORT QUESTION, TO CONCLUDE

The grid solved the particle in milliseconds; the network matched it and then recovered the diffusivity from noisy readings in the same run. For which questions about a battery would you train a network, and for which solve a grid?

# Key Takeaways

## THE CELL MODEL

|                       |                                  |
|-----------------------|----------------------------------|
| <b>what it is</b>     | five fields on two length scales |
| <b>function</b>       | the voltage from the fields      |
| <b>train, tune</b>    | pick the level by the current    |
| <b>vs one physics</b> | each field feeds the others      |
| <b>used for</b>       | what a BMS cannot measure        |

## HEAT AND THE INVERSE

|                          |                                    |
|--------------------------|------------------------------------|
| <b>what it is</b>        | a particle PINN, start built in    |
| <b>function</b>          | the field, and $D_s$ from readings |
| <b>train, tune</b>       | scaled units; balance in the loss  |
| <b>vs finite volumes</b> | equal accuracy, plus the inverse   |
| <b>used for</b>          | the parameter that ages most       |

*Next — L10.2: the same transport and charge, in a ceramic cell at 800 °C.*

# Questions

## 1 – 2 · A NETWORK FOR THE PARTICLE

- 1** How can a network learn the concentration of lithium inside the particle along the radius and in time?

The network takes the radius and the time as inputs and is trained until the diffusion equation holds at every point, without solved data. The initial concentration and the symmetry about the centre are known exactly, so they are built into the form of the solution rather than learned. *Liu ch. 2; Ex\_10.1 section 5*

- 2** Why must the radius, the time and the concentration be scaled before the network is trained?

In SI units these quantities differ by many orders of magnitude, and the gradient of the loss is dominated by the largest terms. Scaled by the particle radius, the discharge time and a reference concentration set by the current, every quantity is of order one and every term is weighted alike. *Ex\_10.1 section 3*

## 3 – 4 · CONSERVATION AND THE INVERSE

- 3** Why does a network not conserve the amount of lithium in the particle by itself, and how is it enforced?

The network satisfies the diffusion equation only approximately, and small errors in the flux at the surface accumulate into lithium gained or lost. The amount that has entered is known from the applied current, so a loss term holds the mean concentration in the particle to that value. *Liu ch. 2; Ex\_10.1 section 5*

- 4** How can the solid-state diffusivity be estimated from measurements at the particle surface, and when is it not possible?

The diffusivity is a trainable parameter, optimised with the network so that the concentration satisfies the diffusion equation and matches the surface measurements. This works when the measurements change with the diffusivity; on the flat voltage plateau of LFP they barely change, and it cannot be identified. *Liu ch. 4; Ex\_10.1 section 6*

NEXT

## L10.2 — Solid Oxide Cells

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The same transport and charge conservation in a ceramic cell at 800 °C, where L9.2's steam channel meets its electrode.

### BEFORE THE NEXT SESSION

|             |                                 |
|-------------|---------------------------------|
| read        | Incropera ch. 14                |
| run         | Ex_10.1, sections 1 to 6        |
| bring       | your diffusivity, and its error |
| think about | what a BMS can and cannot see   |

*Play the film at 1C and at 5C before the session, and watch what the voltage does on the LFP plateau.*